# Asthma *Prediction* Protocol

| | | |
|---|---|---|
| 191<br>USABLE PATIENTS | 15,691<br>PATIENT-DAYS | 144<br>EXACERBATION DAYS · 0.92 % |

## §1 Research objectives

Two questions, kept in separate tables and never mixed. Both are asked of one patient-day at a time. **Change the window and horizon** the ablation holds the question fixed and shrinks the input.

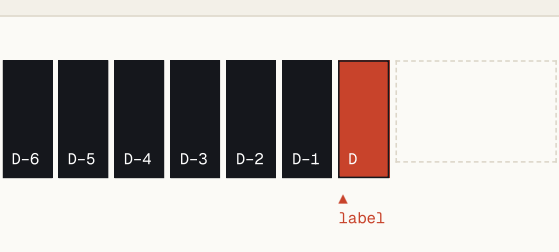
input window · --days X 7 6 5 4 3 2 1

onset horizon · --horizon H within 3 within 2 tomorrow only

TIMELINE A · DETECTION

### Is day D an exacerbation day?

| | |
|---|---|
| input | X days up to and including day D — `--days X` |
| output | probability that day D is an exacerbation day |
| asked on | 15,691 days — 144 positive (0.92 %) |
| ablation | Input — 7 days in, then 6, 5, 4, 3, 2, down to day D alone |



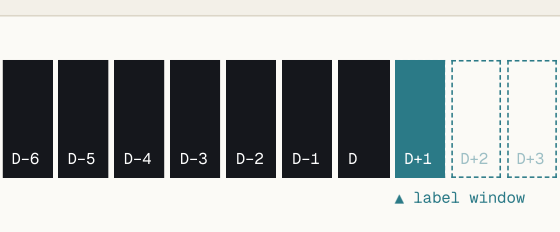
*X = 7 days in* → model →

P(day D is an exacerbation day)

TIMELINE B · ONSET

### Will a new exacerbation start on day D+1?

| | |
|---|---|
| input | X days up to and including day D — identical to detection |
| output | probability that a new episode starts within H days after D |
| asked on | 15,196 days — 151 positive (0.99 %) at the labels' 1–3 day horizon |
| two axes | X shrinks 7→1 as above, *and* the question tightens: within 3 days, within 2, or tomorrow only |



*X = 7 days in* → model →

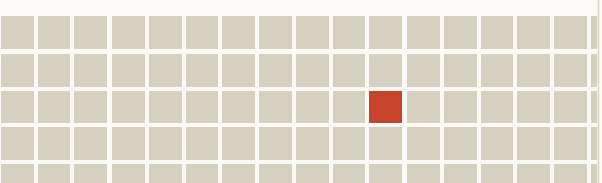
P(episode starts on day D+1) · --horizon 1

## §2 Study population

| | | |
|---|---|---|
| AIIMS ADULTS<br>110<br>patients | AIIMS PAEDIATRIC<br>54<br>patients | COPD<br>27<br>patients |

TOTAL USABLE
191
38 with exacerbation · 153 without

DAYS — ONE IN 109 IS POSITIVE (TO SCALE)



| | | |
|---|---|---|
| 15,691<br>patient-days | 144<br>exacerbation days · 0.92 % | 15,547<br>normal days |

PATIENTS → DAYS

| | |
|---|---|
| with exacerbation · 38 | 4,976 d |
| without · 153 | |
| total · 191 | 15,691 d |

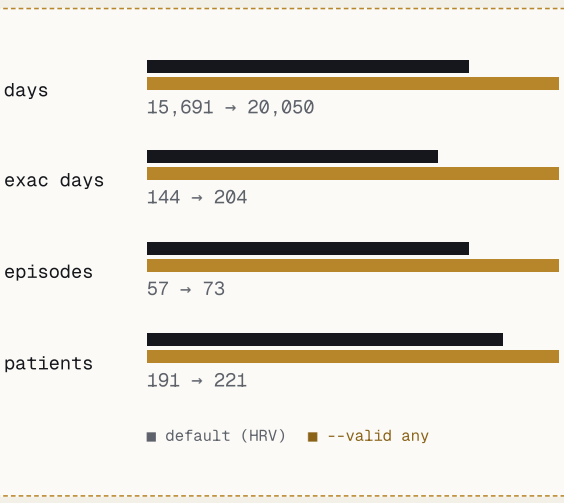
DISTRIBUTION BY GROUP

| group | patients | days | exac days | exac pts |
|---|---|---|---|---|
| aiims_adult | 110 | 9,338 | 125 | 26 |
| aiims_paed | 54 | 4,981 | 8 | 6 |
| copd | 27 | 1,372 | 11 | 6 |

ELIGIBILITY · AN UNUSED OPTION WORTH 42 % MORE POSITIVES

The study's `sample` column means *"the day has at least one HRV reading"*, so days with heart rate but no HRV are marked unscoreable — and so are the episodes they belong to. Those days carry a median of **710 heart-rate readings each**. Widening the rule to *has HR or HRV* is built as `join --valid any` — opt-in and never the default, because switching changes every number in the paper.

**We will obviously RUN this experiment later**



### Available measurements

coverage over 40 sampled patients

**Every model gets every variable, always.** The question is never "can HRV alone predict this" — it is "can these measurements predict this". A path decides what happens to HR and HRV; every other variable goes in raw alongside.

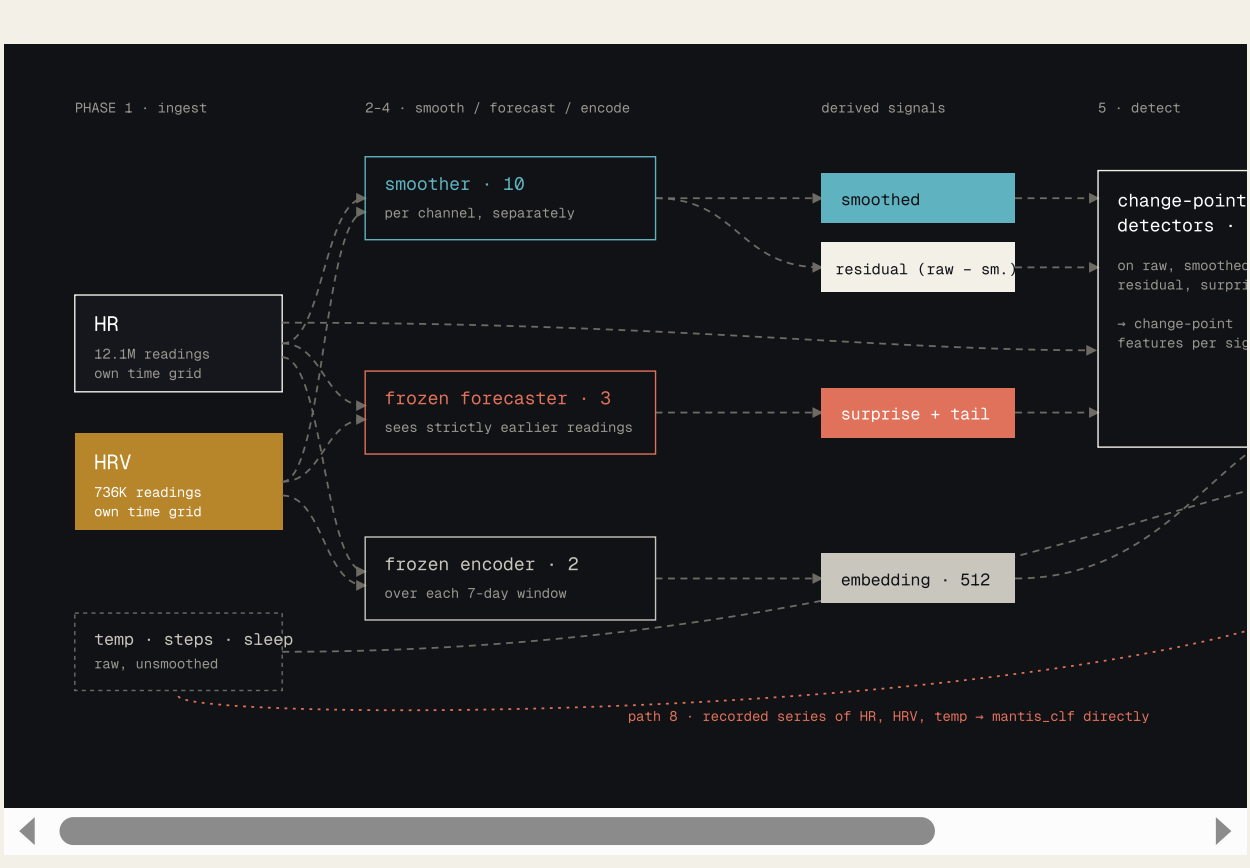
| VARIABLE | COVERAGE | WHAT THE MODEL GETS PER DAY |
|---|---|---|
| `HR` | 100 % | average, spread, lowest, highest — plus smoothed and residual on the smoothing paths |
| `HRV` | 6.1 % | the same |
| `temp` | 8.0 % | average, spread, lowest, highest |
| `steps` | 2.8 % | the same |
| `spo2` | 21 total | left out — 21 readings in the whole cohort. |
| `sleep_active` | 100 % | minutes asleep, and how many separate bouts |
| `steps_active` | 100 % | minutes active, and how many bouts |
| `deep_sleep light_sleep almost_awake` | per bout | total minutes that day, and the bout count |

SLEEP STAGES The export gives one row per sleep bout with the minutes already totalled — for example 45 min deep, 20 min light, 0 almost awake. **It never says when deep sleep started or ended.** So no timeline is assumed and bout lengths are never averaged: a day gets the totals and the bout count, nothing more.

## §3 Signal processing

raw · smoothing · residual · surprise · CPD · embeddings

Each channel branches into several derived signals. Every branch is a **phase** a stage that runs **once**, over the whole cohort, and stores what it produced. The paths then pick from the shelf; nothing is recomputed per path, per model or per fold.

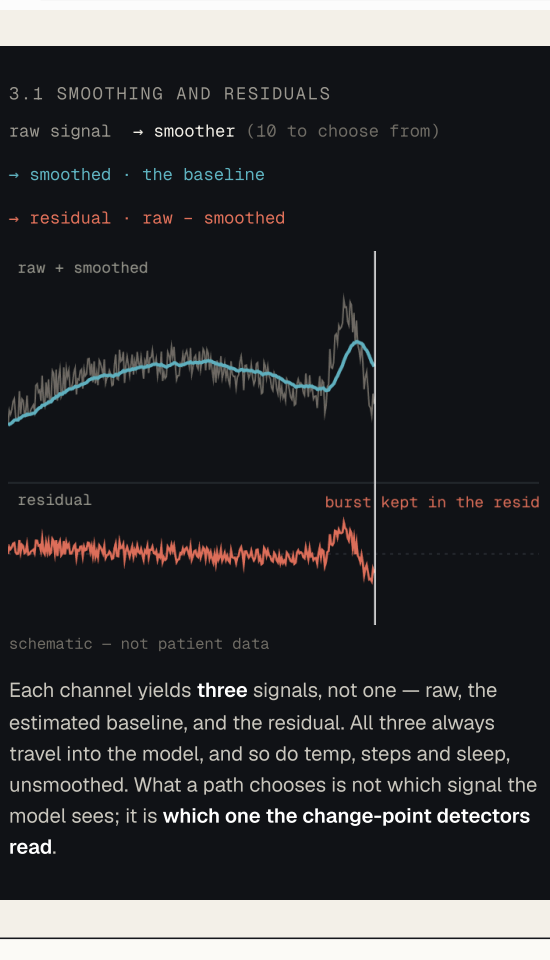


3.1 SMOOTHING AND RESIDUALS

raw signal → smoother (10 to choose from)

→ smoothed · the baseline

→ residual · raw − smoothed



schematic — not patient data

Each channel yields **three** signals, not one — raw, the estimated baseline, and the residual. All three always travel into the model, and so do temp, steps and sleep, unsmoothed. What a path chooses is not which signal the model sees; it is **which one the change-point detectors read**.

3.2 SMOOTHER SCORES · HRV ROUND 1, ALL PATIENTS

| METHOD | S JOINT | R | K | LVL ERR | LOOK. |
|---|---|---|---|---|---|
| kim | 0.585 | 0.528 | 0.647 | 5.30% | 0.11 |
| krlst | 0.580 | 0.647 | 0.520 | 5.06% | 0.27 |
| **rspf** | 0.578 | 0.456 | 0.733 | 0.54% | 3.16 |
| gpssm | 0.573 | 0.410 | 0.800 | 4.26% | 0.33 |
| ctimm | 0.555 | 0.415 | 0.743 | 3.13% | 0.10 |
| hrkf | 0.552 | 0.425 | 0.716 | 2.37% | 0.36 |
| anrewma | 0.534 | 0.596 | 0.479 | 2.67% | 0.00 |
| logmhw | 0.505 | 0.693 | 0.369 | 5.34% | 1.73 |
| gammadglm | 0.456 | 0.305 | 0.681 | 2.15% | 0.13 |
| ossa | 0.389 | 0.175 | 0.866 | 2.04% | 0.74 |

Round-1 lead: rspf — lowest level error (0.54 %), S within 1.2 % of the best. Full analysis → §3.5.

3.2A EACH METRIC, ITS FORMULA, EVERY SMOOTHER

HRV · round 1 · all patients · x = raw, x̂ = smoothed

S joint | R remove | K keep | Level error | Lookahead

### S — joint score

`S = √( R × K )`

One number that rewards removing noise and keeping the signal at the same time. Zero if either side is zero, so both useless smoothers score 0.

better: **higher ↑**

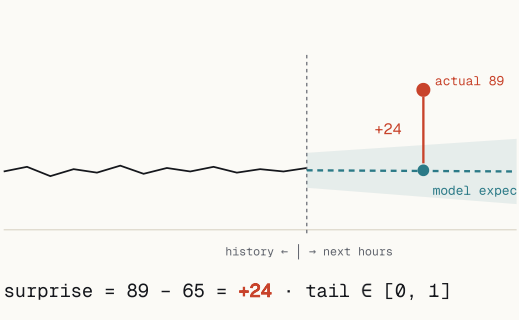
useless-smoother check: do nothing → 0 · flat line → 0

| | | | |
|---|---|---|---|
| 1 | kim | √(0.528 × 0.647) | 0.585 |
| 2 | krlst | √(0.647 × 0.520) | 0.580 |
| 3 | **rspf** | √(0.456 × 0.733) | 0.578 |
| 4 | gpssm | √(0.410 × 0.800) | 0.573 |
| 5 | ctimm | √(0.415 × 0.743) | 0.555 |
| 6 | hrkf | √(0.425 × 0.716) | 0.552 |
| 7 | anrewma | √(0.596 × 0.479) | 0.534 |
| 8 | logmhw | √(0.693 × 0.369) | 0.505 |
| 9 | gammadglm | √(0.305 × 0.681) | 0.456 |
| 10 | ossa | √(0.175 × 0.866) | 0.389 |

The four base scores behind R and K (variance, roughness, peak preservation, correlation) and residual std are defined in Table 5.3; per-smoother values for those appear in the radar figure `signal_radar.png` (§6.7).

3.3 SURPRISE — FORECAST ERROR, FROM A FROZEN MODEL

Both 3.3 and 3.4 replace a hand-built quantity with something a pretrained model computes; neither is trained on this cohort. A frozen forecaster is shown the history (~3.5 days) and asked what comes next (the next hours). The surprise is how wrong it was.



surprise = 89 − 65 = **+24** · tail ∈ [0, 1]

A smoother would *absorb* this — it is built to track a drifting baseline, so after two or three high readings its baseline rises and the residual shrinks. A forecaster predicting from before the change does not. It also reports where 89 fell in its own predictive distribution (**tail**, 0 to 1), which is comparable across patients in a way "+24" is not: 24 means different things at a 30 and a 90 baseline.

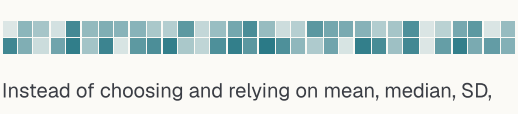
**The only clean residual in the pipeline.** Frozen weights, nothing fitted on the cohort, each block predicted from strictly earlier readings — so unlike nine of the ten smoothers there is no lookahead to argue about.

3.4 EMBEDDING — LEARNED REPRESENTATION

7-day window →

frozen encoder · MOMENT / MantisV2 →

512 numbers / day → model



Instead of choosing and relying on mean, median, SD, min and max, the encoder learns a representation of the **entire temporal pattern**. The 512 numbers are reduced inside the training fold, never before it — reducing earlier would fit on test patients.

THE CONTEXT BUDGET · DAYS × PROFILE = INPUT LENGTH

1 2 3 4 5 6 7

512 slots · 7 day(s) ≈ **19.7 min / slot**

MOMENT is fixed at 512, MantisV2 wants a multiple of 32 and was pretrained at 512, Timestral takes up to 32,768. So against a fixed ceiling, changing `--days` does not change how much the model is handed — it changes the **resolution** those slots are spent at. 7 days into 512 is ~20 minutes a slot; 1 day is ~2.8 minutes. HRV's 7-day window is 385 points and fits natively; HR's is 5,208 and needs `--profile 73`. `windowing.bins_for(7, 512)` returns 73, so the arithmetic is never implicit.

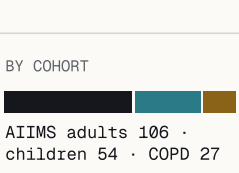
3.5 HRV SMOOTHER SELECTION · ROUND 1

29 Oct 2026 · @jatin · signal metrics of §5.3

### In the first round of scoring, rspf came out ahead — not yet the final choice.

PATIENTS SCORED
187
the `smoothable` tier · 38 had an exacerbation

BY COHORT



AIIMS adults 106 · children 54 · COPD 27

HRV SIGNAL
SDNN · ~1 / 10 min
capped at 129 by the device

SETTINGS DERIVED FROM DATA
3 sets
AIIMS · COPD · all patients combined. Methods doc: link to add

WHY ONE SCORE IS NOT ENOUGH

Only level error can be used alone. Every other score is maxed by a useless smoother.

| USELESS SMOOTHER | PERFECT ON | ZERO ON |
|---|---|---|
| Do nothing (x̂ = x) | correlation, peak preservation | variance and roughness reduction |
| Flat line (x̂ = the average) | variance and roughness reduction | correlation, peak preservation |

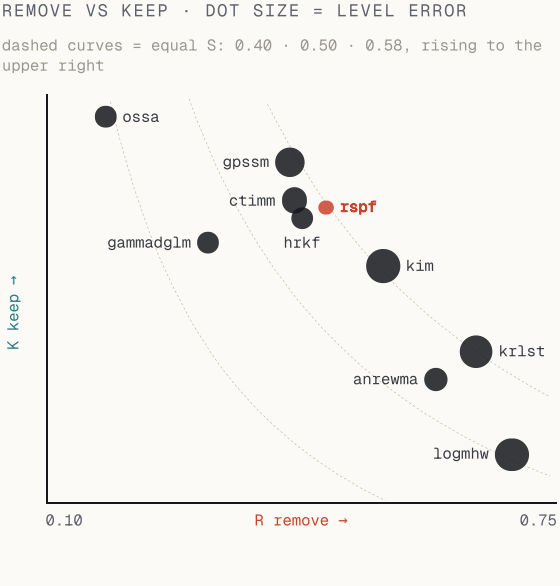
So scores are combined in pairs that pull against each other (reductions as fractions, 0 to 1):

```
R remove = √( variance_reduction ×
roughness_reduction )
K keep   = √( peak_preservation ×
correlation )
S joint  = √( R × K )
```

A square root of a product is 0 if either part is 0, so both useless smoothers score 0. Level error is kept separate: shifting the level never buys anything, so it should not be traded against the others.

REMOVE VS KEEP · DOT SIZE = LEVEL ERROR

dashed curves = equal S: 0.40 · 0.50 · 0.58, rising to the upper right



| METHOD | R REMOVE | K KEEP | S JOINT | LEVEL ERROR % · LOWER ↓ |
|---|---|---|---|---|
| kim | 0.528 | 0.647 | 0.585 | 5.30 |
| krlst | 0.647 | 0.520 | 0.580 | 5.06 |
| **rspf** | 0.456 | 0.733 | 0.578 | 0.54 |
| gpssm | 0.410 | 0.800 | 0.573 | 4.26 |
| ctimm | 0.415 | 0.743 | 0.555 | 3.13 |
| hrkf | 0.425 | 0.716 | 0.552 | 2.37 |
| anrewma | 0.596 | 0.479 | 0.534 | 2.67 |
| logmhw | 0.693 | 0.369 | 0.505 | 5.34 |
| gammadglm | 0.305 | 0.681 | 0.456 | 2.15 |
| ossa | 0.175 | 0.866 | 0.389 | 2.04 |

sorted by S · all patients combined

#### 1 · S can't separate the top 4

kim, krlst, rspf, gpssm are within 1.2 % of each other, and the top spot changes by cohort.

#### 2 · Level error separates them clearly

rspf has the lowest level error in every cohort (0.37–0.59 %). No other method gets below 1.30 % in any cohort: 3 to 4 times worse.

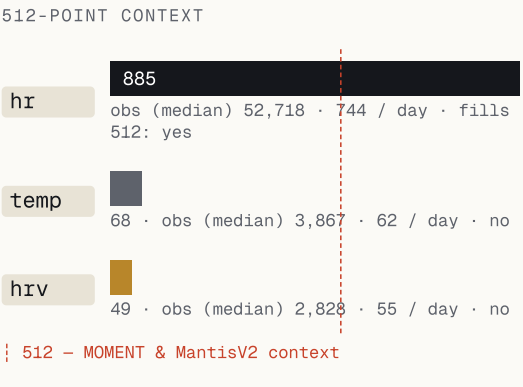
#### 3 · rspf loses almost nothing for it

Its S is only 1.2 % below the best and its K is 3rd highest, so it still smooths properly.

**Radar patterns** (every score on 0–1; outward = more of that property, not better; on mean error outward is worse). Over-smoothing — `logmhw`, `krlst` : big on variance and roughness, small on peak. Barely smoothing — `ossa` : big on peak, small on variance and roughness.

**Next step.** These scores only compare smoothed data to raw data — they do not tell us which smoother helps predict exacerbations. With exacerbation-day labels now available, each smoother's output goes into the prediction models; the one that improves prediction and detection becomes final for the real-time pipeline (figure , §6.7). For that round: kim, krlst, gpssm and ctimm scored close to rspf and are worth testing too; ossa (barely smooths) and logmhw (loses peaks) look weakest so far; rspf fits each patient's settings on the whole recording, so for real-time use its settings must be fitted only on past data (§3.2).

3.6 READINGS PER UNBROKEN SEGMENT VS A 512-POINT CONTEXT



A foundation model needs its context *contiguous*, and MOMENT and MantisV2 both want 512 points. A 49-reading HRV segment cannot fill that; an 885-reading HR segment can. **This is why the foundation routes run on HR.**

3.7 PHASES — BUILD STATUS

1 **ingest** yes
Read the export, lay each channel on its own time grid, write the labels. → all 10 channels, one file each

2 **smooth** 10 × 2 channels yes
Run the 10 smoothers on HR, then on HRV — separately, same pipeline. → smoothed + residual, per channel

3 **forecast** 3 × 2 built, not run
Run the 3 frozen forecasters on HR and HRV. → surprise · tail

4 **encode** 2 built, not run
Run the 2 frozen encoders over each 7-day window. → an embedding per patient-day

5 **detect** 4 per signal yes
Run the 4 detectors on each signal the earlier phases produced — raw, smoothed, residual, surprise. → change-point features per signal

6 **join → model → score** once per path yes
Assemble a path's ingredients into one table per patient-day, split the patients 80:20, train, score. → one row per path, model and fold

## §4 Experimental design

paths · classifiers · 80:20 splits · 5 seeds · ablations

One path is one row of the comparison: an input, the models, a probability. **Every path feeds the model all the variables** — the path only decides what happens to HR and HRV. **Click a row** for its rationale.

| PATH | HR/HRV raw | smoothed | residual | surprise + tail | embedding | series (no summary) | temp·steps·sleep | always_no | xgboost | catboost | rnn | lstm | mantis_clf | FOUNDATION-MODEL AXIS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| PATH 1 · baseline<br>Raw baseline | ● | | | | | | ● | ● | ● | ● | ● | ● | | – |
| PATH 2 · smoothing alone<br>Smoothed | ● | ● | | | | | ● | ● | ● | ● | ● | ● | | – |
| PATH 3 · change points alone<br>Raw + change points | ◆ | | | | | | ● | ● | ● | ● | ● | ● | | – |
| PATH 4 · smoothed<br>Smoothed + change points | ● | ◆ | | | | | ● | ● | ● | ● | ● | ● | | – |
| PATH 5 · main hypothesis<br>Residual + change points | ● | ● | ◆ | | | | ● | ● | ● | ● | ● | ● | | – |
| PATH 6 · tsfm<br>Surprise + change points | ● | | | ◆ | | | ● | ● | ● | ● | ● | ● | | timestral · timesfm · chronos (3 |
| PATH 7 · tsfm<br>Embedding | ● | | | | ● | | ● | ● | ● | ● | ● | ● | | moment · mantis (2 × 5) |
| PATH 8 · the open question<br>MantisV2 as the model | | | | | | ● | ● | ○ | ○ | | | | ● | mantis_clf · full / head |

● fed to the model ◆ change points computed on this signal ● classifier run ○ the row to beat / reference

IMPLEMENTATION FLOW · PATH 5 · BUILT · RUN

### Residual + change points

↻ replay

1 · INGEST — HR · HRV · temp · steps · sleep — per patient, per day
2 · SMOOTH — HR & HRV → baseline — 10 smoothers, per patient
3 · RESIDUAL — raw − baseline — drift removed, bursts kept
4 · CHANGE POINTS — detect regime shifts — on the residuals
5 · SUMMARISE — average · spread · lowest · highest — over the X-day window
6 · JOIN — one table per patient-day — + temp · steps · sleep · activity
7 · CLASSI — alway · xgboo · catbo · rnn · lstm — 80:20 patient split

WHAT NEVER CHANGES, IN ANY PHASE OR PATH

| | |
|---|---|
| the patients | All 191. A patient a method could not handle gets blank numbers, not a removed day. |
| the split | 80:20, whole patients to train or test and never both — the same patients every time. **5 seeds.** |
| the variables | All of them reach the model, in every path. |
| the question | Detection and onset stay in separate tables. |
| the scoring | The same metrics, counted per day, per episode, and per probability band. |

The run asserts this: if two rows in a comparison were ever scored on different days, different positives or different patients, it stops rather than report them.

train · 80 % · 153 patients | test · 38

whole patients · stratified by group · × seeds 0–4

ABLATIONS & AXES

**Input window** · X = 7, 6, 5, 4, 3, 2, 1 days — on both tasks. `sweep window`

**Onset horizon** · within 3 days, within 2, tomorrow only. Same days, narrower label — the denominator never moves. `--horizon`

**Foundation model** · on paths 6 and 7 a second axis: each foundation model produces its own features and all five classifiers read them — three forecasters × five classifiers on path 6, two encoders × five on path 7. Each runs **one** foundation model at a time, so the comparison varies the model and nothing else.

**Mantis fine-tuning** · `fine_tuning_type=full` trains the whole body (the experiment), `head` freezes it (the conservative reading) — both keyed, so they compare.

## §5 Evaluation protocol

confusion matrix · metrics · formulas · test counts · intervals

Seven numbers judge a prediction, and six more judge a smoother on its own. Nothing else is reported.

THE FOUR COUNTS EVERYTHING IS BUILT FROM

A day is **flagged** when its predicted probability is **0.5 or above**. Every threshold metric below is one question about these four numbers.

| | was an exacerbation day | was normal |
|---|---|---|
| flagged | TP<br>caught — exacerbation day correctly flagged | FP<br>false alarm — normal day incorrectly flagged |
| not flagged | FN<br>missed — exacerbation day missed | TN<br>left alone — normal day correctly identified |

| | |
|---|---|
| N | total evaluated days |
| p | predicted probability of exacerbation |
| y | actual outcome: 1 = exacerbation, 0 = normal |

INTERVALS

### 95 % intervals come from resampling *test patients* — not days.

Days within a patient are dependent, and resampling them gives an interval several times too narrow. The five primary metrics carry an interval; lead time and detection delay are medians over caught episodes and are reported as point estimates.

The bootstrap is ~85 % of a full grid's wall clock (1000 draws: 2.9 s per evaluation). `--boot 0` skips it and is 145× quicker — iterate with it, turn it on once for the final tables.

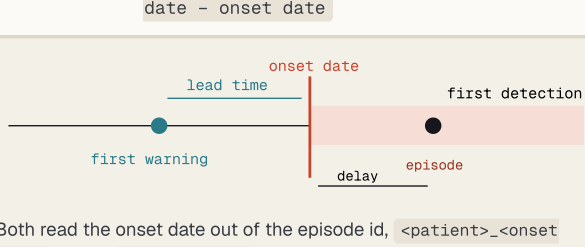
### Table 5.1 · Primary model performance

| METRIC | WHAT IT MEANS IN OUR EXPERIMENT | FORMULA | BETTER | INTERVAL |
|---|---|---|---|---|
| **AUROC** | How well does the model rank exacerbation days above normal days? | area under the ROC curve | higher ↑ | yes |
| **Sensitivity** | Out of all exacerbation days, how many did the model correctly identify? | TP / (TP + FN) | higher ↑ | yes |
| **Specificity** | Out of all normal days, how many did the model correctly identify? | TN / (TN + FP) | higher ↑ | yes |
| **PPV** | Out of all flagged days, how many were actually exacerbation days? | TP / (TP + FP) | higher ↑ | yes |
| **Brier score** | How close are the predicted probabilities to the actual outcomes? | (1/N) ∑ (p − y)² | lower ↓ | yes |

AUROC is threshold-free — it only cares about the ordering, so it does not move if the 0.5 cut moves. Brier is the only one that judges the *probability* rather than the ranking. The other three all change with the threshold.

### Table 5.2 · Clinical usefulness

| METRIC | MEANING · FORMULA | BETTER | READ ON |
|---|---|---|---|
| **Lead time** | How many days before an exacerbation begins does the model first warn us? `onset date − first warning date` | higher ↑ | onset |
| **Detection delay** | How many days after an exacerbation begins does the model first identify it? `first detection date − onset date` | lower ↓ | detection |



Both read the onset date out of the episode id, `<patient>_<onset date>` — e.g. `a001_20251203` . Exact mirrors, so **lead + delay = 0** on any set of predictions; the selftest asserts it. Reported as the median across caught episodes; undefined when nothing was caught.

### Table 5.3 · Signal metrics — smoothers only

| METRIC | WHAT IT MEANS | FORMULA | BETT |
|---|---|---|---|
| mean_error_pct | level error: how far, in %, the smoother moved the average. Up or down both count. The only score usable alone | 100 · \|mean(x̂) − mean(x)\| / \|mean(x)\| | lowe · 0 idea |
| variance_reduction | how much of the spread was removed. Not always better: 100 % is a flat line with no information | 100 · (1 − Var(x̂) / Var(x)) | high more remo |
| roughness_reduction | how much jumpiness between back-to-back readings was removed. 100 % is again a flat line | rough(x) = mean \|xᵢ₊₁ − xᵢ\| · 100 · (1 − rough(x̂) / rough(x)) | high smoo |
| peak_preservation | how much of each raw burst survives, 0–1. Doing nothing also scores 1, so read it with the two above | median over peaks of clip(smoothed rise / raw rise, 0, 1) | high |
| residual_std | size of what the smoother removed, in HRV units. Matters because change detection runs on this residual | std(x − x̂) | not scor = noth remo |
| correlation | how closely smoothed follows raw, −1 to 1. Pearson r: straight-line co-movement, blind to a level shift. Spearman ρ: Pearson on ranks. Doing nothing scores 1 | Pearson r · Spearman ρ | high |

These judge a smoother directly against the raw trace, with no model and no labels. Averaged per patient and then across patients, so one long recording cannot dominate — and they are the only scores available for a patient with no exacerbation record. Notation: x = the raw HRV reading, x̂ = the smoothed value a method gives.

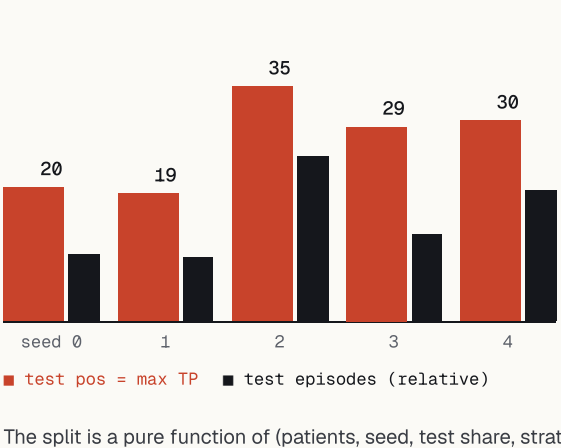
**Peak preservation, step by step — the most important check.** If an exacerbation shows up as a burst in HRV, a smoother that flattens bursts can score well on variance and roughness but is useless for early warning. Physiologically we don't know what is pathological for HRV, so we keep away from assuming smoothed HRV is a better predictor than raw. (1) Find peaks p in the raw signal: points that stand out by at least half the patient's own spread. (2) Look 6 readings before and after each (about 2 hours). (3) Rise = peak value minus the lowest value in that window, for raw and for smoothed. (4) ratio$_p$ = clip(smoothed rise / raw rise, 0, 1): 1 = kept fully, 0 = flattened. (5) Take the median over all peaks, so one huge spike can't dominate.

Variance asks "how far from the average?"; roughness asks "how far from the previous reading?". A slow drift has high variance but low roughness; fast jitter is the opposite. Round-1 scores → §3.5.

5.4 TEST SAMPLE COUNTS · DETERMINISTIC SPLIT, PER SEED

detection | onset

| patients | patient-days | positive days | normal days | base rate | episodes |
|---|---|---|---|---|---|
| 191 | 15,691 | 144 | 15,547 | 0.92 % | 57 |



| seed | train days | train pos | train eps | test days | test pos = max TP | test eps |
|---|---|---|---|---|---|---|
| 0 | 12,889 | 124 | 47 | 2,802 | 20 | 10 |
| 1 | 12,593 | 125 | 47 | 3,098 | 19 | 10 |
| 2 | 12,669 | 109 | 43 | 3,022 | 35 | 14 |
| 3 | 12,875 | 115 | 48 | 2,816 | 29 | 9 |
| 4 | 12,984 | 114 | 44 | 2,707 | 30 | 13 |

The split is a pure function of (patients, seed, test share, strata), so these numbers are **deterministic** — they will come out the same on any machine. The positive count is the ceiling on TP: a model cannot catch more exacerbation days than the test set holds. 38 of the 191 patients ever have an exacerbation; 80:20 puts 38 patients in test and 153 in train. Seed-to-seed variation → §7 limitations.

### 5.5 What gets compared — fill in from the run

one row per path × architecture · per task · per seed

| PATH | ARCHITECTURE | TP | FP | TN | FN | AUROC | SENS | SPEC | PPV | BRIER |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 raw | always_no | – | – | – | – | – | – | – | – | – |
| 1 raw | xgboost | – | – | – | – | – | – | – | – | – |
| 1 raw | catboost · rnn · lstm | – | – | – | – | – | – | – | – | – |
| 4 smoothed+cpd | all 5 classifiers | – | – | – | – | – | – | – | – | – |
| 5 residual+cpd | all 5 classifiers | – | – | – | – | – | – | – | – | – |
| 6 surprise+cpd | × timestral | – | – | – | – | – | – | – | – | – |
| 6 surprise+cpd | × timesfm | – | – | – | – | – | – | – | – | – |
| 6 surprise+cpd | × chronos | – | – | – | – | – | – | – | – | – |
| 7 embedding | × moment | – | – | – | – | – | – | – | – | – |
| 7 embedding | × mantis | – | – | – | – | – | – | – | – | – |
| 8 tsfm as model | mantis_clf | – | – | – | – | – | – | – | – | – |

The raw rows land in `data/reports/experiments/runs_<task>.csv` and the averaged ones in `summary_<task>.csv` . `always_no` is the row that fixes what every other number means. Rows 6 and 7 multiply out across the foundation-model axis (§4).

## §6 Results

detection · onset · cross-path · signal metrics

One block per path; inside it the two tasks; inside each task the **seven metrics of §5**, each as an ablation curve. All curves share the same x-axis — **days of input, 1 to 7** — so any metric reads across paths. **No interpretation is offered here.** The method is in §1–4, the metrics in §5, the denominators in every caption. The reader judges.

**Read against this.** Six paths × two tasks × seven metrics × seven window lengths is several hundred numbers, against about **10 test episodes per fold**. Some will separate by chance. Pre-register one primary comparison before running, and report how many were made.

metric AUROC Sensitivity Specificity PPV Brier Delay / Lead

DETECTION

15,691 days · 144 positive · 0.92 %

### AUROC vs days of input · all paths

AUROC

awaiting run · path{n}_detection_auroc.png

1 2 3 4 5 6 7 days

path 1 path 4 path 5 path 6 path 7 path 8

Shaded band = 95 % interval, from resampling test patients.

ONSET

15,196 valid days · 151 positive · --horizon 1

### AUROC vs days of input · all paths

AUROC

awaiting run · path{n}_onset_auroc.png

1 2 3 4 5 6 7 days

path 1 path 4 path 5 path 6 path 7 path 8

Shaded band = 95 % interval, from resampling test patients.

### 6.1–6.6 · Per-path figure slots

Six slots per task: five primary metrics on every task, plus the one clinical metric that applies — **detection delay** on detection, **lead time** on onset. 72 slots across the six paths. Hover a slot for its filename.

| | DETECTION · 15,691 DAYS · 144 POSITIVE · BASE RATE 0.92 % | | | | | | ONSET · 15,196 VALID DAYS · 151 POS | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | AUROC | Sens | Spec | PPV | Brier | Delay | AUROC | Sens | Spec | PP |
| 6.1 · PATH 1<br>Raw baseline<br>built · run | 1–7d | 1–7d | 1–7d | 1–7d | 1–7d | 1–7d | 1–7d | 1–7d | 1–7d | 1–7d |
| 6.2 · PATH 4<br>Smoothed + change points<br>built · run | 1–7d | 1–7d | 1–7d | 1–7d | 1–7d | 1–7d | 1–7d | 1–7d | 1–7d | 1–7d |
| 6.3 · PATH 5<br>Residual + change points · main hypothesis<br>built · run | 1–7d | 1–7d | 1–7d | 1–7d | 1–7d | 1–7d | 1–7d | 1–7d | 1–7d | 1–7d |
| 6.4 · PATH 6<br>Surprise + change points · tsfm<br>built · not run | 1–7d | 1–7d | 1–7d | 1–7d | 1–7d | 1–7d | 1–7d | 1–7d | 1–7d | 1–7d |
| 6.5 · PATH 7<br>Embedding · tsfm<br>built · not run | 1–7d | 1–7d | 1–7d | 1–7d | 1–7d | 1–7d | 1–7d | 1–7d | 1–7d | 1–7d |
| 6.6 · PATH 8<br>MantisV2 as the model · the open question<br>built · not run | 1–7d | 1–7d | 1–7d | 1–7d | 1–7d | 1–7d | 1–7d | 1–7d | 1–7d | 1–7d |

Caption pattern: **{Metric} vs days of input · path {n} · {task}.** Primary: "Shaded band = 95 % interval." Clinical: "Median over caught episodes; point estimate." Filename:

`path{n}_{task}_{auroc|sensitivity|specificity|ppv|brier|detection_delay|lead_days}.png`

```
# How to add a figure: save it to
data/reports/figures/ under the filename the
slot names, then replace the dashed placeholder
with one line. Keep the <figcaption>. Figures
are published alongside the page, so the src
stays a plain relative path.
<img src="figures/path1_detection_auroc.png"
  alt="Path 1 detection AUROC vs days of input">
```

### 6.7 Across paths, and the smoothers on their own

Once a window length is settled from the curves above, these compare the paths at that one setting. The signal metrics judge a smoother against the raw trace, with no model and no labels involved.

paths_detection.png

**Detection, all paths** — at the chosen window. One row per path × model; identical test patients in every row, asserted at run time.

paths_onset.png

**Onset, all paths** — at --horizon 1.

calibration.png

**Calibration** — predicted against observed rate across 10 probability bands. Brier compresses this to one number; the curve shows where it goes wrong.

by_group.png

**By group** — 125 exacerbation days among the adults against 8 among the children, so the paediatric interval will be wide.

cohorts.png

**Tier 1 vs 2 vs 3** — Tier 3's base rate is 3.1× tier 1's, so its numbers are not comparable in absolute terms.

signal_mean_hrv.png

**Mean HRV** — dashed line = raw mean; each dot = one smoother's mean.

signal_radar.png

**All scores at once** — one radar per smoother, every score on 0–1. Outward = more of that property, not better; on mean error outward is worse.

smoothers_model.png

**Ten smoothers** — through the same model and folds. anrewma is the only one causal end to end.

signal_correlation.png

**Correlation** — raw against smoothed. Higher better.

signal_variance_reduction.png

**Variance reduction %** — variance removed. Higher better.

signal_roughness_reduction.png

**Roughness reduction %** — jitter removed. Higher better.

signal_peak_preservation.png

**Peak preservation** — 0–1. Bursts kept while jitter removed.

signal_residual_std.png

**Residual std** — spread of raw − smoothed. Lower better.

signal_mean_error_pct.png

**Mean error %** — departure from the raw trace. Lower better.

## §7 Discussion

findings · sensitivity analyses · limitations · implications

No results from this cohort are interpreted here (§6). What follows is the reasoning behind the design and the evidence it rests on, each point linked to where it is shown.

7.1 FINDINGS · WHY A FOUNDATION MODEL IS NOT SIMPLY USED AS THE PREDICTOR

The obvious move is to hand the raw series to a large pretrained model and let it predict. That is not the main approach here, for reasons worth stating because they will be asked.

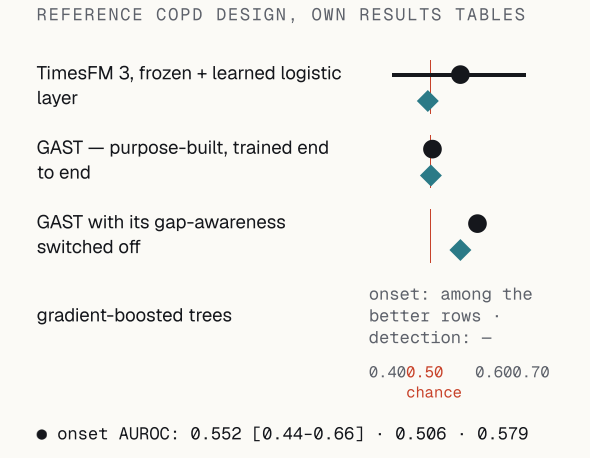
### Pretraining transfers the wrong knowledge

Pretraining teaches a model **what time series look like** — trends, rhythms, bursts, regime changes. It teaches it nothing about **what an asthma exacerbation looks like**. That mapping has to be learned from the labels, and there are **144 positive days** (§2). So the job splits by where the information comes from: signal knowledge from the big frozen model, which needs no labels; disease knowledge from a small classifier, which must learn a 144-example mapping and does so better for being small.

### "Direct" is mostly the same thing in disguise

144 positives cannot fine-tune a 110M–385M parameter body. So "use MOMENT directly" necessarily means freeze the body and train a small head — which *is* the embedding route (§3.4), with the head restricted to one linear layer. Running embeddings into `xgboost` is strictly more general.

THE NEAREST EVIDENCE SAYS IT LOSES · REFERENCE COPD DESIGN, OWN RESULTS TABLES

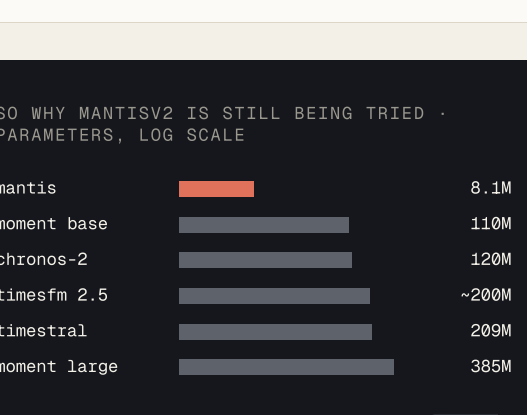


● onset AUROC: 0.552 [0.44–0.66] · 0.506 · 0.579

◆ detection AUROC: 0.499 · 0.502 · 0.553

The frozen foundation model landed at exactly chance on detection. The purpose-built end-to-end network did no better, and was beaten by its own ablation.

SO WHY MANTISV2 IS STILL BEING TRIED · PARAMETERS, LOG SCALE



**8.1M parameters.** The only model on the shortlist in the same universe as 144 positives. MOMENT-large is 47× bigger. End-to-end training here is a real option rather than memorisation.

**A native classifier.** `fit(X, y)` / `predict_proba(X)` — not a head bolted onto a forecaster.

**Cross-channel structure.** It takes several channels at once and can *learn a combination of them.* Frozen per-channel embeddings cannot: each channel is embedded in ignorance of the others. This is the one capability the other routes genuinely lack.

**Near-zero cost.** Apache-2.0, CPU-feasible, one file. The result is informative either way, and a clean negative answers the question readers will ask.

Not Timestral, TimesFM or Chronos-2 for this: they only forecast. No classification head, and nothing exposed to attach one to. → path 8, §4

## 7.2 SENSITIVITY ANALYSES

### Tier 2 · smoothable

Excludes 4 patients carrying **6 days and 0 positives.** State it arithmetically or run it with `--stratify False`, because restricting the universe moves a patient across the split boundary in 3 of 5 seeds and a patient carries ~71 days. Evidence: §2 populations

### Tier 3 · exacerbators

Selects on the outcome, inflating the base rate 3.1× and making the false-alarm rate unmeasurable in the population that would be deployed to — a different question, never the headline. Its numbers are not comparable in absolute terms. Evidence: §2, figure `cohorts.png`

### Wider eligibility

`join --valid any` would add 42 % more positives (204 exacerbation days, 73 episodes, 221 patients), but switching changes every number in the paper, and onset validity would have to be rederived. Evidence: §2 eligibility

## 7.3 LIMITATIONS

### Test sets vary between seeds

**The test positives nearly double between seeds** — 19 on seed 1 against 35 on seed 2 for detection, 23 against 41 for onset. Whole patients move, and an exacerbator carries all of their days with them. So part of the spread between seeds is the test set being easier or harder, not the model changing. This is why a result is read as a mean across the five with its interval, never from one seed, and why the run asserts that every model in a fold saw *identical* test days, positives, episodes and patients. Evidence: §5.4

### Small number of episodes

57 scoreable episodes, about 10 per test set, against several hundred numbers — some will separate by chance. Pre-register one primary comparison. The paediatric group has 8 exacerbation days against 125 among the adults, so the paediatric interval will be wide. Evidence: §2, §6

### Lookahead in nine smoothers

Nine of ten smoothers set per-patient settings from the whole recording; residuals from them are optimistically biased for onset. Only anrewma (and the frozen-forecaster surprise) is clean. The round-1 leader, **rspf**, has the largest truncation deviation of all ten (3.16); for real-time use its settings must be fitted only on past data. Evidence: §3.2–3.3, §3.5

### Smoother choice is label-free so far

Round 1 compares smoothed to raw only; it does not show which smoother helps predict exacerbations. The top four by S (kim, krlst, rspf, gpssm) sit within 1.2 % and swap by cohort, so the final pick waits for the model round. Evidence: §3.5

### Sparse channels

HRV covers 6.1 %, temp 8.0 %, steps 2.8 %; spo2 is left out with 21 readings in the whole cohort. HRV segments (49 readings) cannot fill a 512-point context, so foundation routes run on HR. Sleep stages carry no timing. Evidence: §2, §3.6

### Build status

As of 30 Oct 2026: paths 1, 4, 5 run; 6, 7, 8 built and not run. `mantis_clf` verified on synthetic data only. Evidence: §4

## 7.4 IMPLICATIONS

**The raw baseline comes first.** Nothing else counts as an improvement until path 1's number exists, and `always_no` fixes what every other number means. §4, §5.5

**Use anrewma or surprise for any onset claim** — a warning is by definition made before the event — or report the others as optimistically biased. §3

**Path 8 is informative either way.** A clean negative answers the question readers will ask; `xgboost` is the row to beat. §4

---

# A Appendix

architectures · smoothers · models · commands · settings

## A.1 Every architecture in the pipeline

■ classifier emits the probability · ■ feature producer, frozen

Eleven architectures, in two groups. A **classifier** emits the probability; a **feature producer** is frozen and emits a signal or a vector that a classifier then reads. Only `mantis_clf` is both.

| ARCHITECTURE | WHAT IT IS | SIZE | ROLE | USED BY |
|---|---|---|---|---|
| `always_no` | never flags a day | – | classifier | every path |
| `xgboost` | gradient-boosted trees, 400 × depth 4, blanks sent down a learned branch | – | classifier | every path |
| `catboost` | oblivious trees with ordered boosting, 400 × depth 4 | – | classifier | every path |
| `rnn` | forward RNN over the day window, hidden 64, 2 layers, read at the classified day | – | classifier | every path |
| `lstm` | the same with LSTM cells — gates that hold a slow drift across the window | – | classifier | every path |
| `mantis_clf` | MantisV2 patch transformer, trained end to end on the recorded series | 8.1M | classifier & foundation model | path 8 |
| `timestral` | Mistral Timestral — 24-layer patch transformer, width 768, patch 32, context to 32,768. Frozen | 209M | forecaster → surprise | path 6 |
| `timesfm` | Google TimesFM 2.5 — patch transformer. Frozen. 3.0 weights are non-commercial | ~200M | forecaster → surprise | path 6 |
| `chronos` | Amazon Chronos-2 — patch transformer with covariate support. Frozen | 120M | forecaster → surprise | path 6 |
| `moment` | MOMENT — T5-family encoder, patch 8, context fixed at 512, takes a missingness mask. Frozen | 110M base · 385M large | encoder → embedding | path 7 |
| `mantis` | MantisV2 as a frozen encoder — the same body as mantis_clf, not trained | 8.1M | encoder → embedding | path 7 |

## A.2 The 10 smoothers

| METHOD | DEVIATION WHEN TRUNCATED | CAUSAL END TO END | R | K |
|---|---|---|---|---|
| rspf | 3.16 | no — whole-recording settings | 0.456 | 0.733 |
| logmhw | 1.73 | no — whole-recording settings | 0.693 | 0.369 |
| ossa | 0.74 | no — whole-recording settings | 0.175 | 0.866 |
| hrkf | 0.36 | no — whole-recording settings | 0.425 | 0.716 |
| gpssm | 0.33 | no — whole-recording settings | 0.410 | 0.800 |
| krlst | 0.27 | no — whole-recording settings | 0.647 | 0.520 |
| gammadglm | 0.13 | no — whole-recording settings | 0.305 | 0.681 |
| kim | 0.11 | no — whole-recording settings | 0.528 | 0.647 |
| ctimm | 0.10 | no — whole-recording settings | 0.415 | 0.743 |
| anrewma | 0 | yes | 0.596 | 0.479 |

Run on HR, then on HRV — separately, same pipeline. `--signal hr|hrv`. Settings derived from the data: one set for AIIMS, one for COPD, one for all patients combined. R/K/S and level error: HRV, round 1 (§3.5). Round-1 lead: **rspf**.

## A.3 Technical settings

| | |
|---|---|
| flag threshold | predicted probability ≥ 0.5 |
| split | 80:20 by whole patient, stratified by group, 5 seeds |
| bootstrap | 1000 draws over test patients · --boot 0 to skip |
| gap rule | a break of 180 minutes or more starts a new segment (chunking.py) |
| window | --days 1–7, shared by join and encode (windowing.py) |
| onset horizon | --horizon 1 \| 2 \| 3 |
| context budget | days × profile = input length · windowing.bins_for(7, 512) → 73 |
| HR window, 7 d | 5,208 points → --profile 73 for a 512 context |
| HRV window, 7 d | 385 points — fits natively; path 8 uses --profile 55 |
| forecast history | ~3.5 days → the next hours |
| embedding | 512 numbers per day, reduced inside the training fold |
| validity | default: day has ≥ 1 HRV reading · --valid any: HR or HRV (detection only) |
| analysis population | --cohort all \| smoothable \| exacerbators |
| mantis fine-tuning | fine_tuning_type = full \| head |

## A.4 Repository orientation

Repository: `asthma-prediction/`. **Run everything from that directory** — paths are relative to it. `python` below means `.venv/bin/python`; call it directly, because activating the venv does not always take in a non-interactive shell.

THE SIX POOLS

Interchangeable methods, discovered from the filesystem. **Adding a method is one file in a pool directory** — there is no list to register in.

| | | |
|---|---|---|
| `smoothing/` | estimate the level at each reading → smoothed + residual | 10 |
| `forecasting/` | predict the next readings → surprise + tail | 3 |
| `cpd/` | flag change points in whichever signal it is given | 4 |
| `encoders/` | turn a window into a vector per patient-day | 2 |
| `models/` | predict the probability | 6 |
| `metrics/` | score a prediction | 19 |

FIVE RULES THE REPO ENFORCES

1 **Steps never import each other.** Only `experiment.py` connects them.

2 **Adding a method is one file.** Pools are read off the filesystem.

3 **Every result records how it was made** — method, settings, inputs, code version — in a `manifest.json` beside it.

4 **Nothing is computed twice.** Results are keyed on a content hash; a matching key is reused.

5 **A comparison varies only the method.** Same test patients, same model, same seed.

```
$ python -m wearable selftest
OK — all invariants hold.
# asserts all five plus every invariant;
touches no data. Run it first.
```

| | | |
|---|---|---|
| `config.py`<br>every constant, path, channel and split setting — the only definitions | `windowing.py`<br>the window rule, shared by join and encode. Never redefine it | `chunking.py`<br>the gap rule: a break of 180 minutes or more starts a new segment |
| `steps/`<br>ingest · smooth · forecast · detect · encode · join · model · evaluate | `experiment.py`<br>the only thing that connects steps: the paths and the sweeps | `selftest.py`<br>what must stay true. Read this to learn what the repo guarantees |
| `data/runs/`<br>every stored result, keyed by content, with its provenance | `data/reports/`<br>regenerable output — tables and figures | |

SENSITIVE

**The patient data is sensitive.** Nothing under `data/` is ever committed — not the raw cohorts, not the stored results, not any patient file.

## A.5 Commands

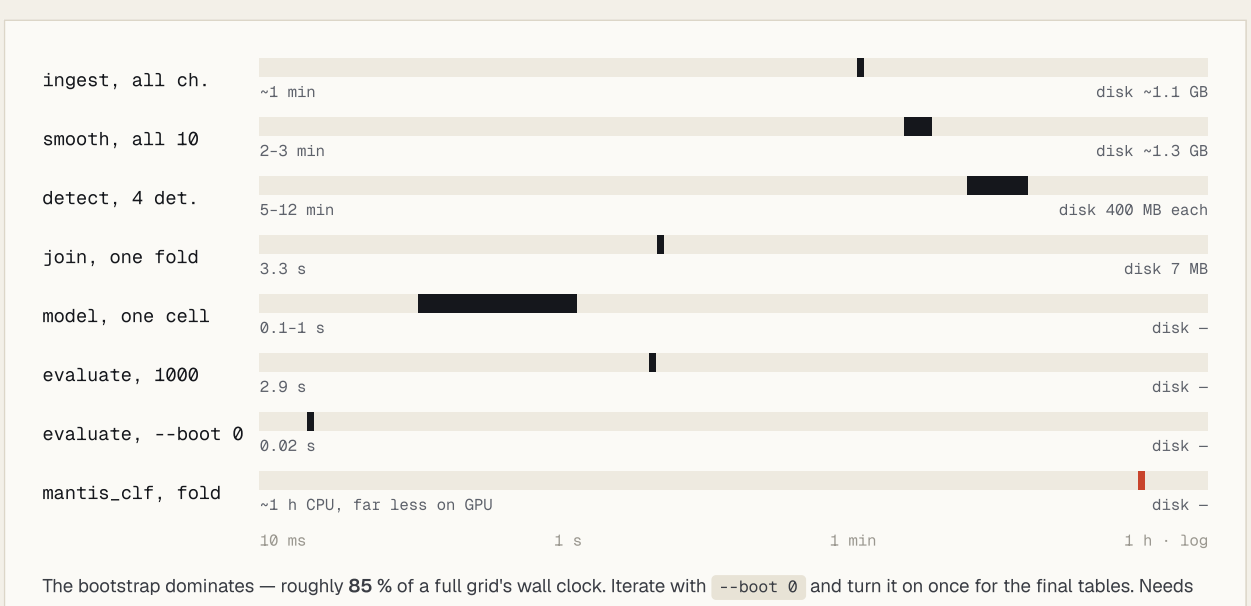
setup · check → compute · experiments

```
# python 3.10; call the venv binary directly
python3.10 -m venv .venv && .venv/bin/pip install -r requirements.txt
.venv/bin/pip install xgboost catboost   # required: selftest fails on any method it cannot import
.venv/bin/pip install mantis-tsfm        # path 8 and the mantis encoder

# optional, opt-in foundation packages
.venv/bin/pip install momentfm
.venv/bin/pip install git+https://github.com/mistralai/timestral.git
.venv/bin/pip install chronos-forecasting "timesfm[torch]"
```

Needs roughly 4 GB free for the stored results and ~2 GB for the packages.

## A.6 Measured cost · 191 patients



The bootstrap dominates — roughly **85 %** of a full grid's wall clock. Iterate with `--boot 0` and turn it on once for the final tables. Needs roughly **4 GB free** for the stored results and ~2 GB for the packages.